07.10.2026

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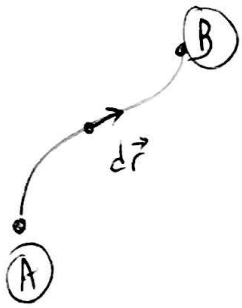
1. Ders

Integral Theorem:

$$\int_a^b g(x)\,dx = f(b) - f(a) \quad \text{where} \quad g(x) = \frac{df(x)}{dx}$$

$$\int_a^b \frac{df(x)}{dx}\,dx = f(b) - f(a)$$

1)

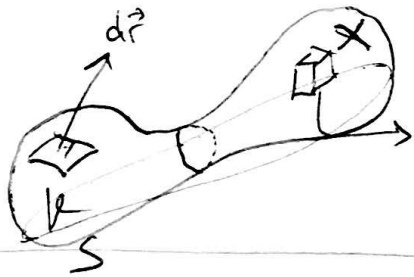


$$d\varphi = (\nabla\varphi)\cdot d\vec{r}$$

$$\int_A^B d\varphi = \varphi(B) - \varphi(A) \quad \propto \quad \int_A^B \nabla\varphi\cdot d\vec{r} = \varphi(B) - \varphi(A)$$

ex: $\vec{F} = -\vec{\nabla}V_F$, $\vec{E} = -\vec{\nabla}V_E$

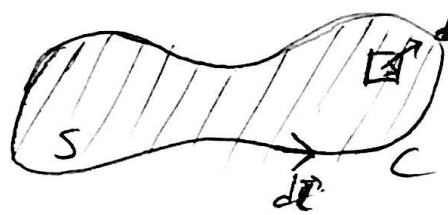
2) $\vec{\nabla}\cdot\vec{A}$



$d\tau$: Volume element. Volume V is enclosed by surface S.

$$\int_V \vec{\nabla}\cdot\vec{A}\,d\tau = \oint_S \vec{A}\cdot d\vec{r}$$ Gauss Theorem

3) $\vec{\nabla}\times\vec{A}$



Surface S is enclosed by closed path C.

$$\int_S \vec{\nabla}\times\vec{A}\cdot d\vec{r} = \oint_C A\cdot dl$$

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Divergence Theorem: Electric Field as an Example.

$$\vec{E} = \frac{q}{4\pi\epsilon_0}\cdot\frac{\hat{r}}{r^2} = \frac{q}{4\pi\epsilon_0}\frac{r\hat{r}}{r^3} = \frac{q}{4\pi\epsilon_0}\frac{\vec{r}}{r^3}$$

Let $\vec{A} = \frac{\vec{r}}{r^3}$, then $\vec{E} = \frac{q}{4\pi\epsilon_0}\vec{A}$, $\boxed{\vec{A} = r^n\vec{r}}$ Generalized for $\vec{E}$, $n=-3$.

We need $\vec{\nabla}\cdot\vec{A}$. Note that $r = (x^2+y^2+z^2)^{\frac{1}{2}}$

then $\frac{\partial r}{\partial x} = \frac{1}{2}\frac{1}{r}2x = \frac{x}{r}$

$$\nabla\cdot\vec{A} = \nabla\cdot\left(x\hat{x}+\hat{y}y+z\hat{z}\right)r^n) = \frac{\partial[xr^n]}{\partial x} + \frac{\partial[yr^n]}{\partial y} + \frac{\partial[zr^n]}{\partial z}$$

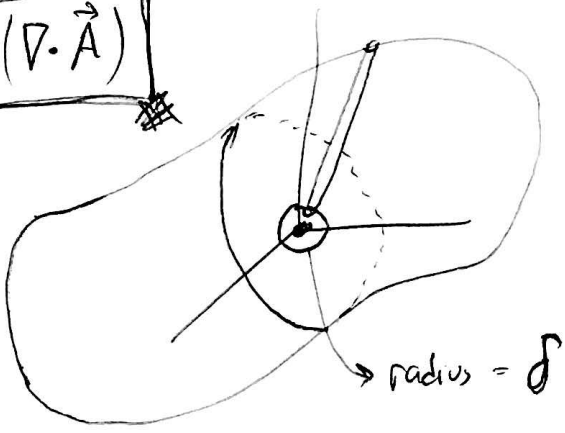
$$= \left(r^n + x\cdot n\cdot r^{n-1}\cdot\frac{\partial r}{\partial x}\right) + (\ldots\ldots) + (\ldots\ldots)$$

$$\nabla\cdot\vec{A} = 3r^n + nr^{n-2}(x^2+y^2+z^2)$$

$$= 3r^n + nr^{n-2}\cdot r^2 = \boxed{r^n(3+n) = (\nabla\cdot\vec{A})}$$

$n=-3$ Case for $\vec{E}$

$$\oint_S \frac{\hat{r}}{r^2}\cdot d\sigma = \int \nabla\cdot\left(\frac{\hat{r}}{r^2}\right)d\tau$$



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There are methods to solve Poisson Eq. wont be covered.

When $\rho = 0$ $\boxed{\nabla^2 \varphi = 0}$ laplace equation

Dirac Delta Function

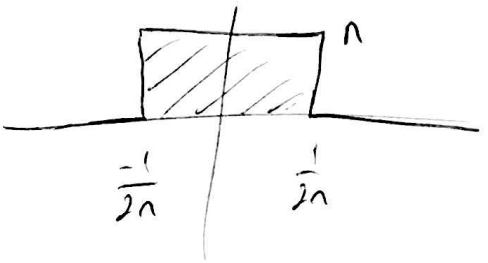
We build a "fuition" with following properties.

i) $\delta(x) = 0 \quad , \quad x \neq 0$

ii) $\int_a^b f(x) \cdot \delta(x) \cdot dx = f(0)$ where $a < 0$ and $b > 0$

$$\Rightarrow \int_{-\epsilon_0}^{\epsilon_0} \delta(x)\, dx = 1$$ can define as: 1) $\delta_n(x) = \begin{cases} n & , |x| < \frac{1}{2n} \\ 0 & , \text{ow} \end{cases}$

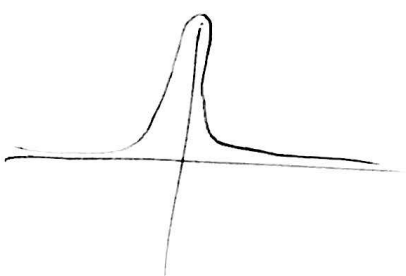
Then $\delta(x) = \lim_{n \to \infty} \delta_n(x)$.



Area $n \cdot \frac{1}{n} = \boxed{1}$

2) $\delta_n(x) = \frac{n}{\sqrt{\pi}} e^{-n^2x^2}$

$$\boxed{\int_{-\infty}^{\infty} \delta_n(x)\, dx = 1}$$



3) $\delta_n = \frac{n}{\pi} \cdot \frac{1}{1 + n^2x^2}$

1. Ders

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$$\lim_{n\to\infty} \int_{-\infty}^{\infty} \delta_n(x) f(x)\, dx = f(0) \quad , \quad \int \delta_n(x)\, dx = 1$$

$$f(x) = 1$$

$$\int_{-\infty}^{\infty} \delta(x) f(x)\, dx = \lim_{n\to\infty} \int_{-\infty}^{\infty} \delta_n(x) f(x)\, dx .$$

Note that $\delta_n(x)$ we have symetric functions.

$$\delta(x) = \delta(-x)$$

Properties of $\delta(x)$.

proof:

1) $\delta(ax) = \frac{1}{a}\, \delta(x) \qquad a > 0$

$u = ax$

1.2) $\int_{-\infty}^{\infty} \delta(ax) \cdot \delta(x)\, dx = \int_{-\infty}^{\infty} \frac{1}{a}\, \delta(x) f(x)\, dx = \frac{f(0)}{a}$

2) $\int_{-\infty}^{\infty} \delta(x) \cdot \delta(x - x')\, dx$, proof $u = x - x'$

$= f(x')$

3) $\int_{-\infty}^{\infty} f(x)\, \delta'(x)\, dx$

$du = \delta'(x)\, dx \qquad u = \delta(x)$

$v = f(x) \qquad dv = f'(x)$

$$= f(x)\, \delta(x) \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \delta(x)\, df(x) \cdot \frac{dx}{dx} = -\int_{-\infty}^{\infty} \delta(x) \cdot f'(x)\, dx$$

$$= -f'(0).$$